

Networks and Systems: MM6

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- **Mm2** Layer-2 Example Technologies: Field-buses CAN and Flexray (hps)
- **Mm3** Clock Synchronisation & impact of clock offsets on sensor data (hps)
- **Mm4 & 5** Network Calculus I & II (HS)
- **Mm6** TCP Modeling and Performance Analysis (hps)
- **Mm7** Performance implications of data access schemes (hps)
- **Mm8** Discrete Event Simulation (HS)
- ...

Layer 4 Performance

2

1) TCP revision / basic introduction

- Connection setup, receiver advertised window, congestion window
- Slow start, congestion avoidance

2) TCP Performance: Simulation and Measurements

- Performance Metrics, TCP over congested router, TCP over WLAN/BT, TCP over GPRS

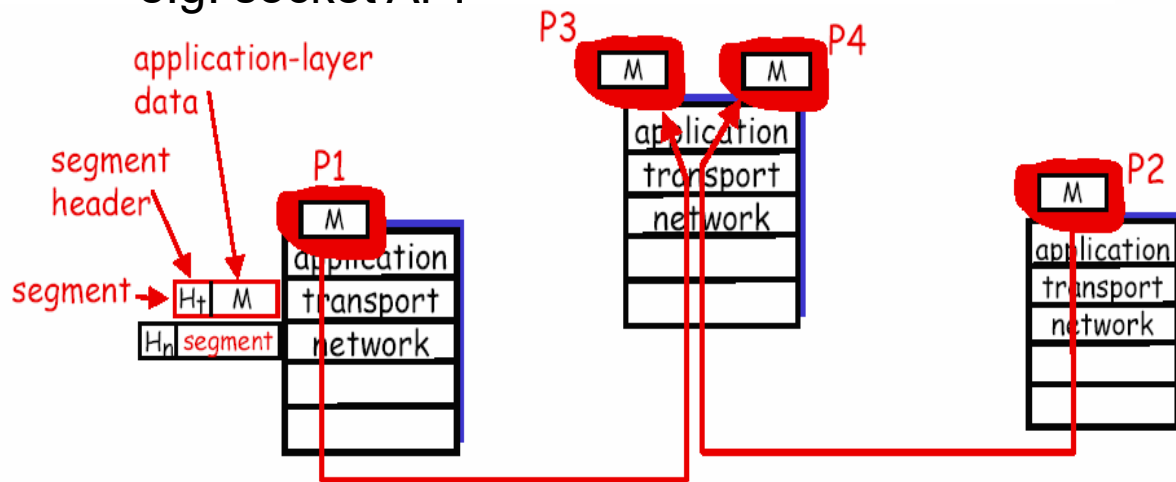
3) TCP Performance Models

- Processor Sharing Models, Fixpoint Models, queuing models with discouraged arrivals

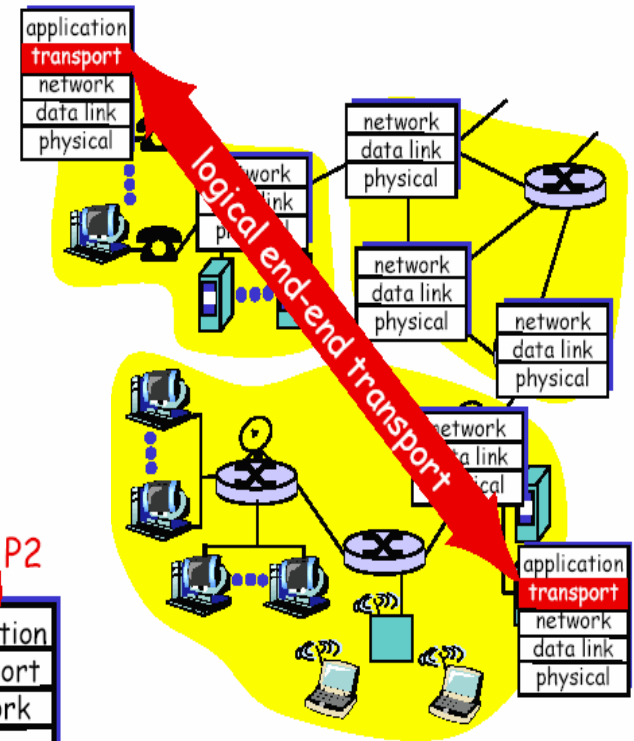
Transport Layer Protocols

Goal: data transfer between application (processes) in end-systems

- support of multiplexing/de-multiplexing
e.g. socket API



data stream/connection identified by:
two IP addresses, protocol number, two port numbers



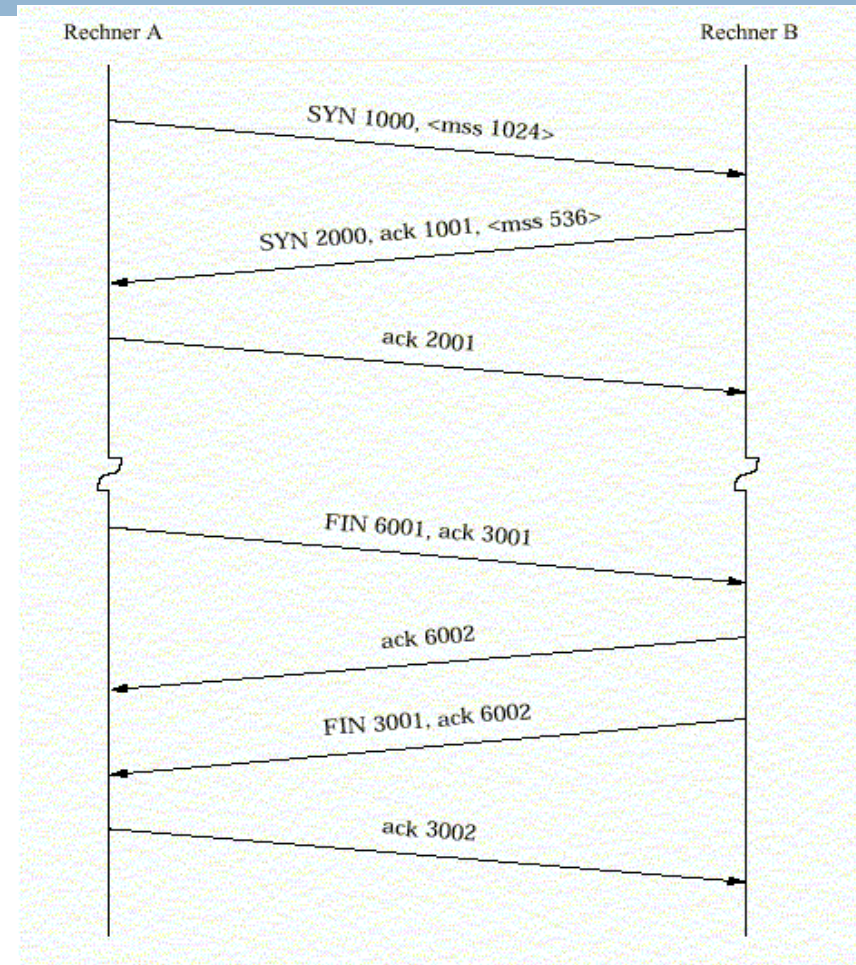
TCP: Connection Establishment / Termination

Connection Establishment:

- 3-way handshake (SYN, SYN-ACK, ACK)
- selection of **initial sequence numbers**
- agreement on **maximum segment size**

Connection end:

- Indicated by FIN flag
- Signalled for both transmission directions
- ,Alternative': RST flag



TCP Flow/Congestion Control: Purpose

□ Flow Control:

- ▣ Adapt sending rate to receiver speed
- ▣ Prevent Buffer Overflow at receiver

Approach: Receiver notifies sender about available buffer space

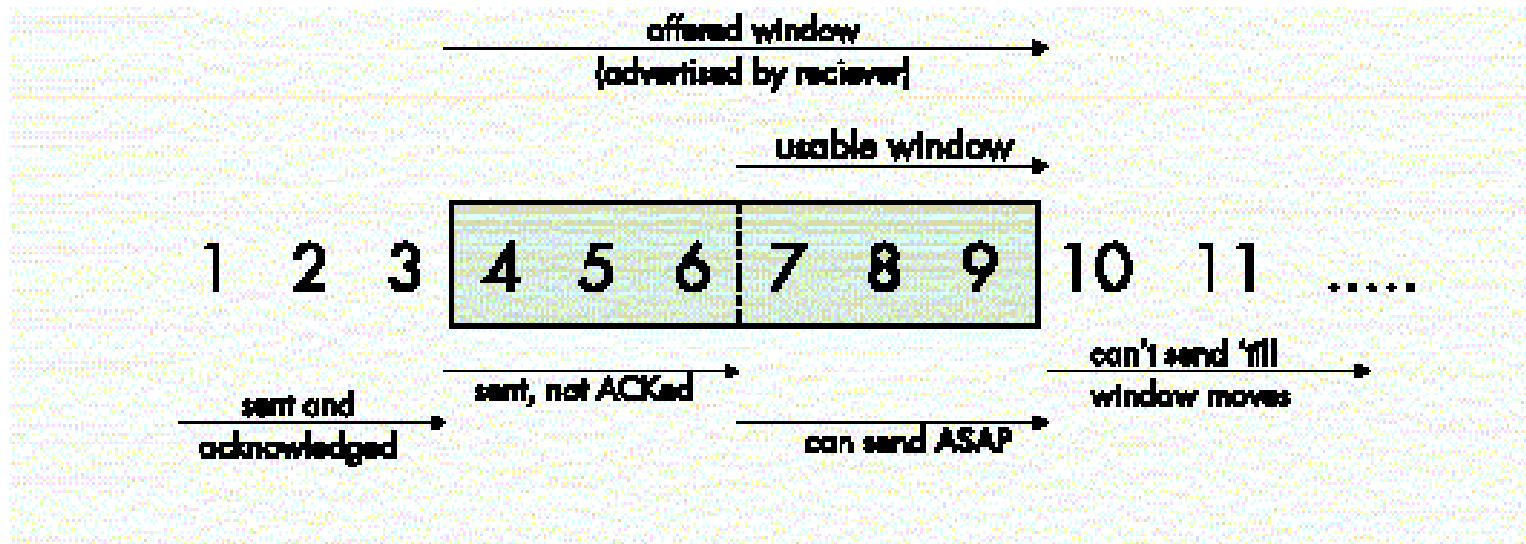
□ Congestion Control

- ▣ Avoid network overload
- ▣ Prevent packet loss at buffers in intermediate systems (routers)
- ▣ ‘fair’ sharing of rates between all TCP connections at bottleneck links

Approach: Congestion Window size adjusted to congestion indicators: timeout, Dup ACK

Approach: ‘Elastic traffic’ – sending rates adjust dynamically when overload indicated

TCP Flow/Congestion Control



- **Sliding Window:** # packets sent and un-acknowledged limited by
 - ▣ receiver buffer (receiver advertised window)
 - ▣ congestion window (purpose: protect intermediate nodes)
- **Congestion window** dynamically adjusted (,probing' for usable bandwidth): increase gradually until congestion indicated via
 - ▣ Timeouts
 - ▣ Duplicate Acknowledgements

TCP Congestion Control:

Slow Start

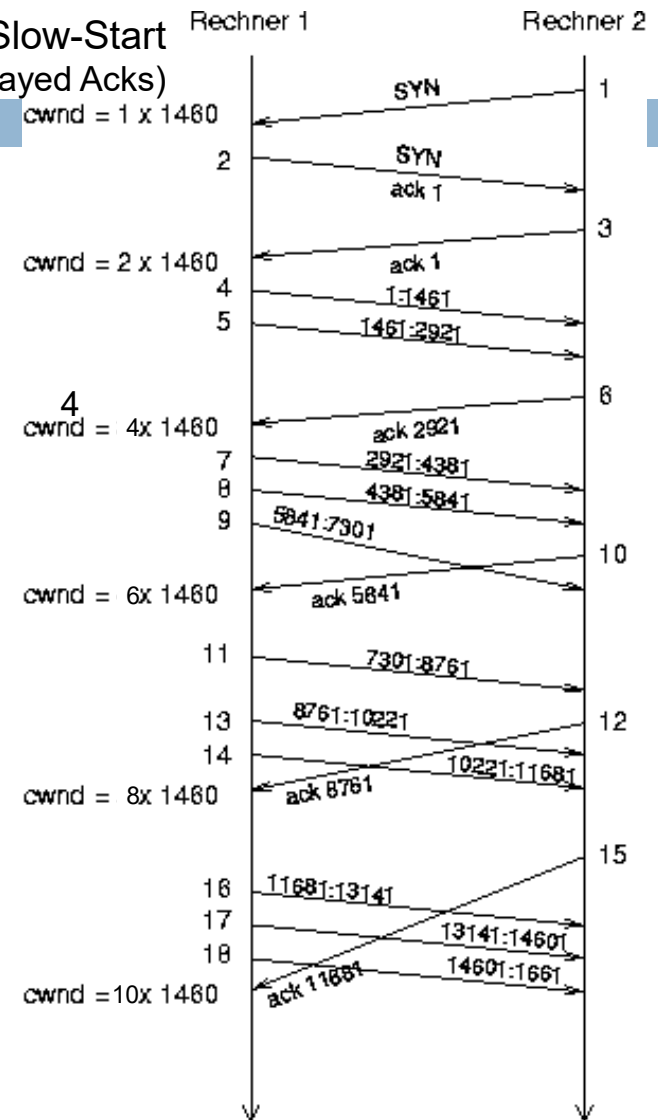
Example: Slow-Start
(receiver using delayed Acks)

Slow-start phase:

- Initial congestion window set to size $cwnd=one$ segment
- Whenever ACK received: Increase $cwnd$ by number of newly acknowledged bytes
- Slow-start phase ends
 - When $cwnd > ssthreshold \rightarrow$ Congestion Avoidance
 - Packet loss detected (timeout, triple dup ACK)

Slow-start initiated after (depends on TCP version)

- connection start
- timeout detection



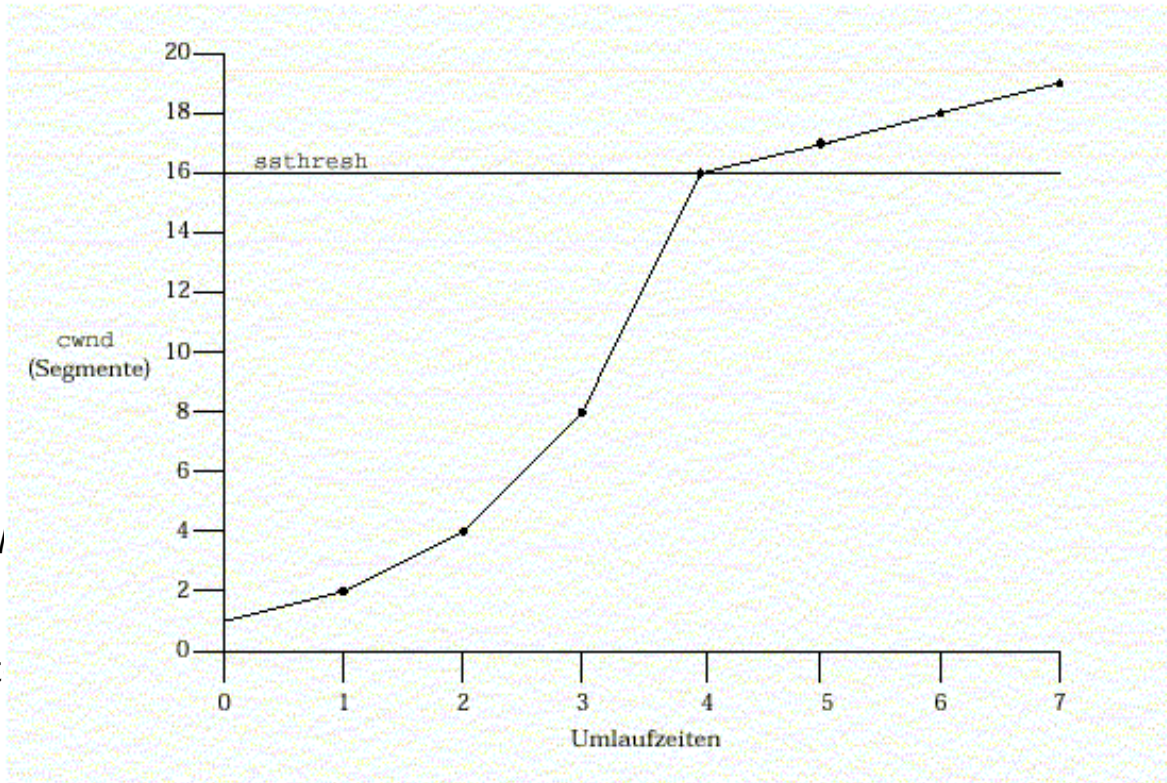
Slow-Start & Congestion Avoidance

For High-Bandwidth Links:

- CWND doubles after each round-trip time (RTT) in slow-start phase
- CWND increases by one segment size in each RTT during congestion avoidance phase

Congestion detected:

- **timeout** → set ssthresh to cwnd, perform slow-start
 - ▣ Time-out value dynamically adjust based on RTT measurements
- **three dup Acks** → half cwnd, ssthresh=cwnd (Fast Retransmit/Recovery)

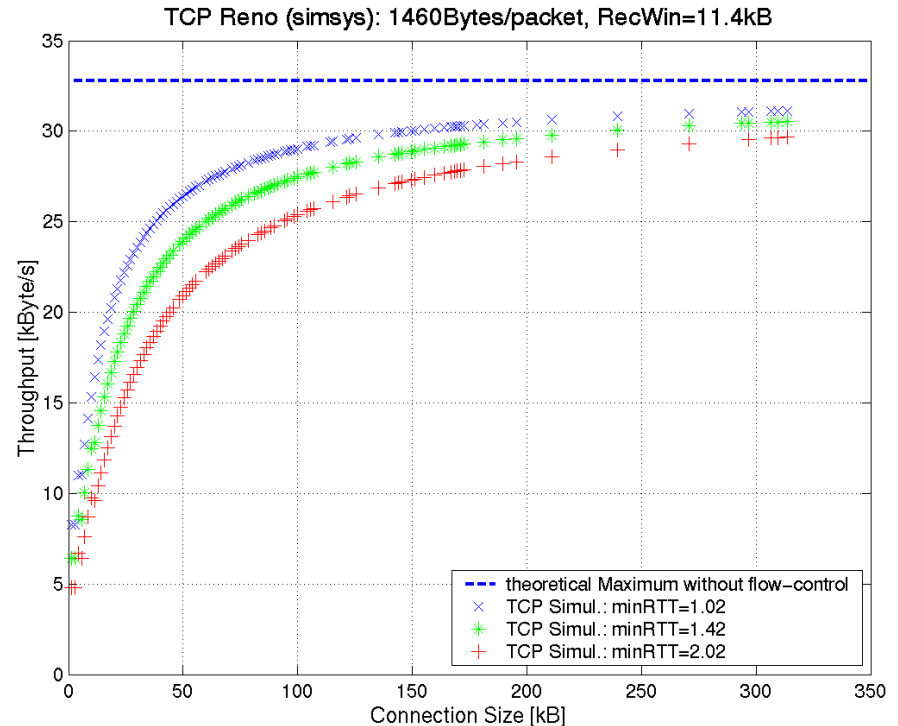


Throughput = $\text{CWND} \times \text{MSS} / \text{RTT}$
(for high-bandwidth links)

Impact of slow-start: Simulation Result

- Larger connection volume → higher average throughput
- Upper Limit:
minimum of Layer-2 throughput and $\text{MaximumWindowSize} / \text{RTT}$

Simulations of GPRS-like
scenario (RTT ~1.2 seconds)
without congestion
(no losses/timeouts/dupACKs)



[Correction:
y-axis in kbit/s]

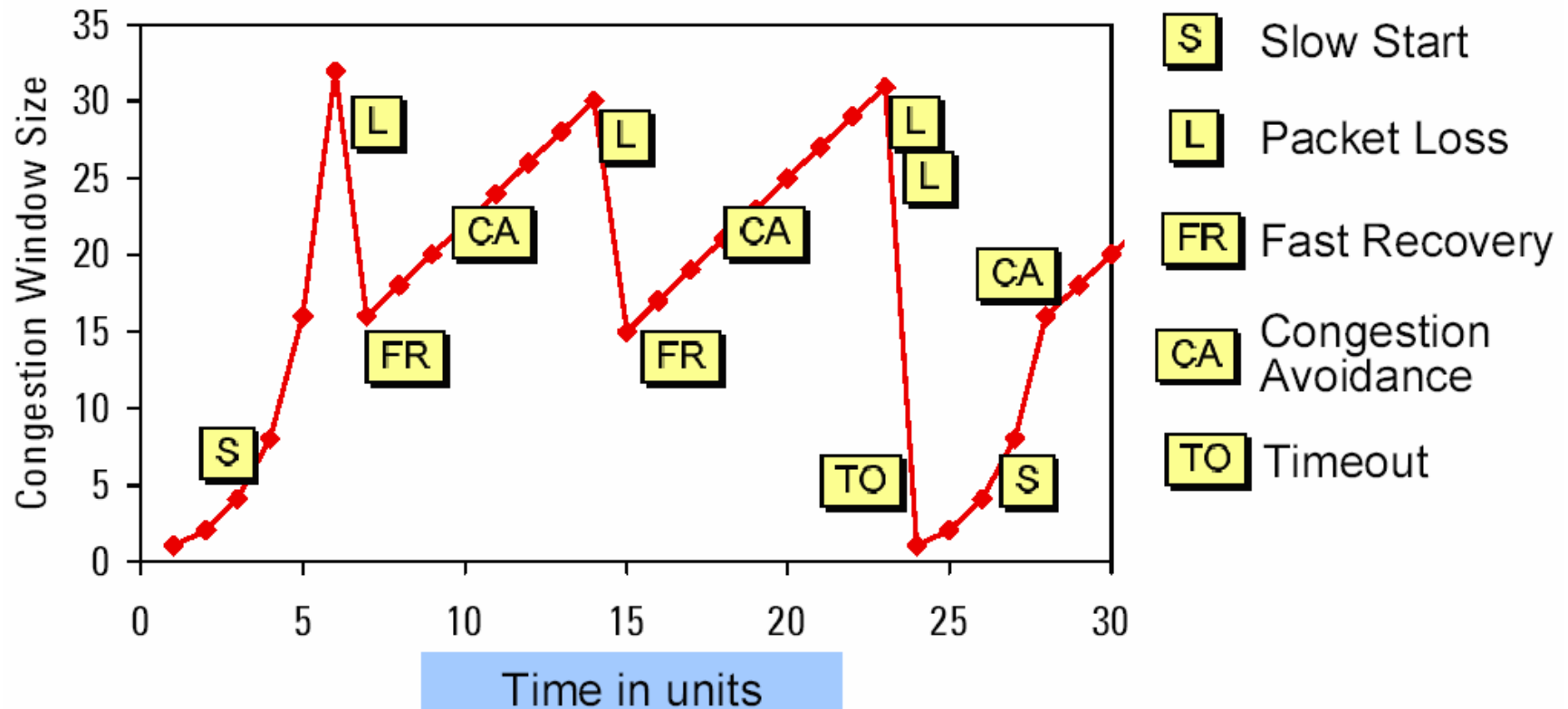
Fast Retransmit

- Goal: Avoid long delays due to waiting for time-outs
- Method: Use indication given by dupAcks, which may be generated due to
 - ▣ packet loss, or
 - ▣ out-of-order packet delivery
- Fast Retransmit:
 - ▣ Upon reception of three **dupAcks** consecutively, assume segment is lost
 - ▣ Retransmit lost packet
 - ▣ Note: 3 dupAcks are also generated if a packet is delivered at least 3 places beyond its in-sequence location

7	9	10	11	8	12
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Illustration of flow control mechanism

■ TCP Reno trace



Layer 4 Performance

12

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TCP Performance Parameters

(Selection)

□ TCP Segment based

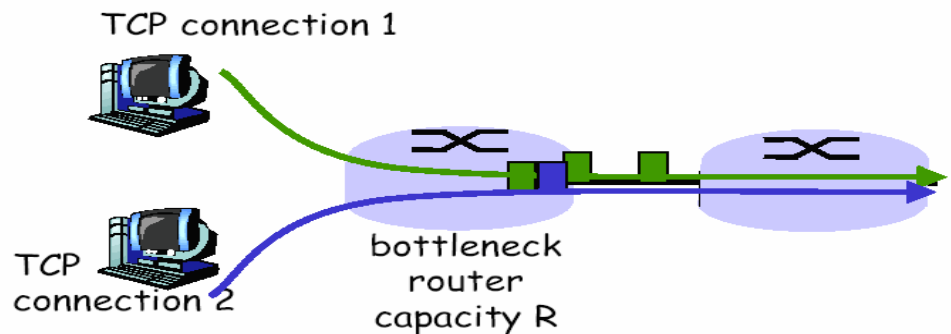
- # of retransmissions, # of time-out events
- Segment RTT delay (max, min, average, distribution)
 - Equals transmission delay + queueing delay + retransmission time(if applicable) for Segment, Waiting Time for Ack sending, ACK delay

□ Connection based

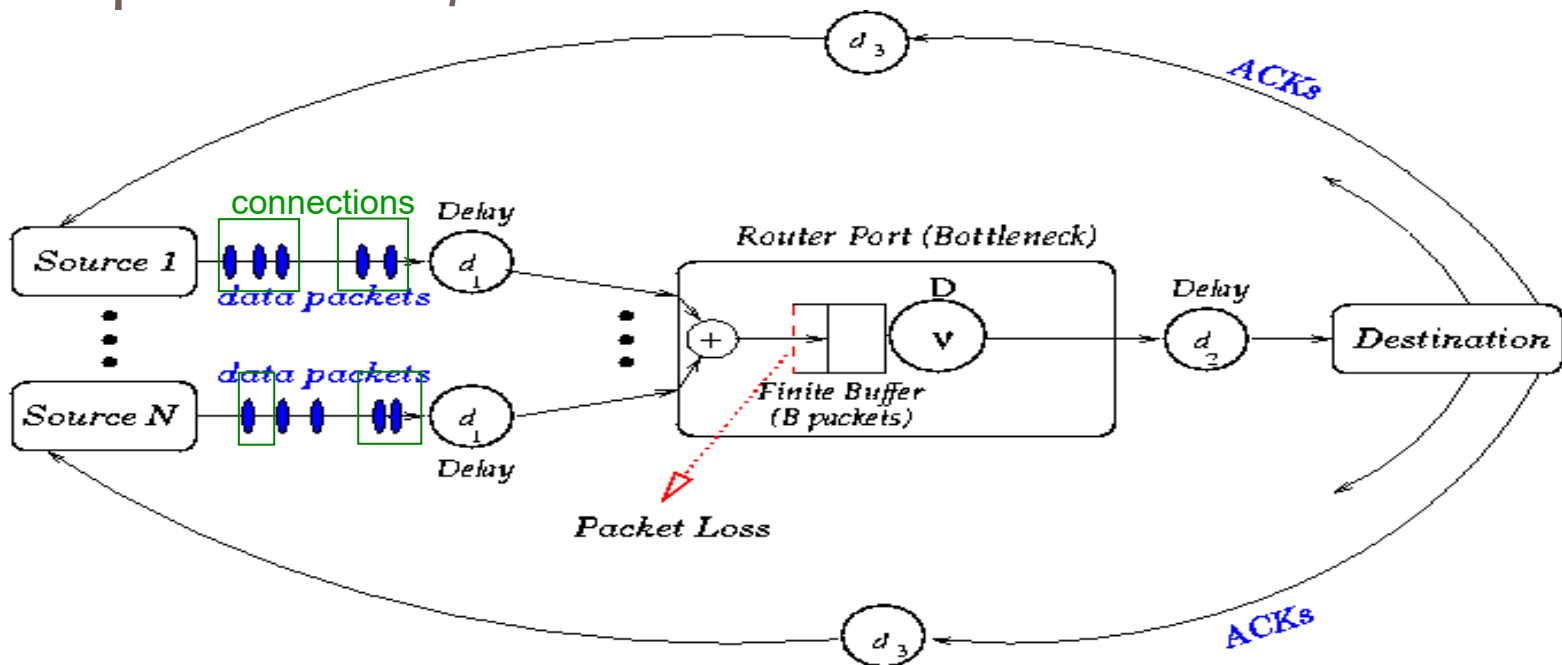
- Data (file) transfer time
- Average Throughput/Goodput
- Probability of successful transfer

□ Fairness:

- Bottleneck scenarios: if N TCP sessions share same **bottleneck** link, each should get $1/N$ of link capacity
- multi-hop scenarios, e.g.
 - **Max-min fairness**

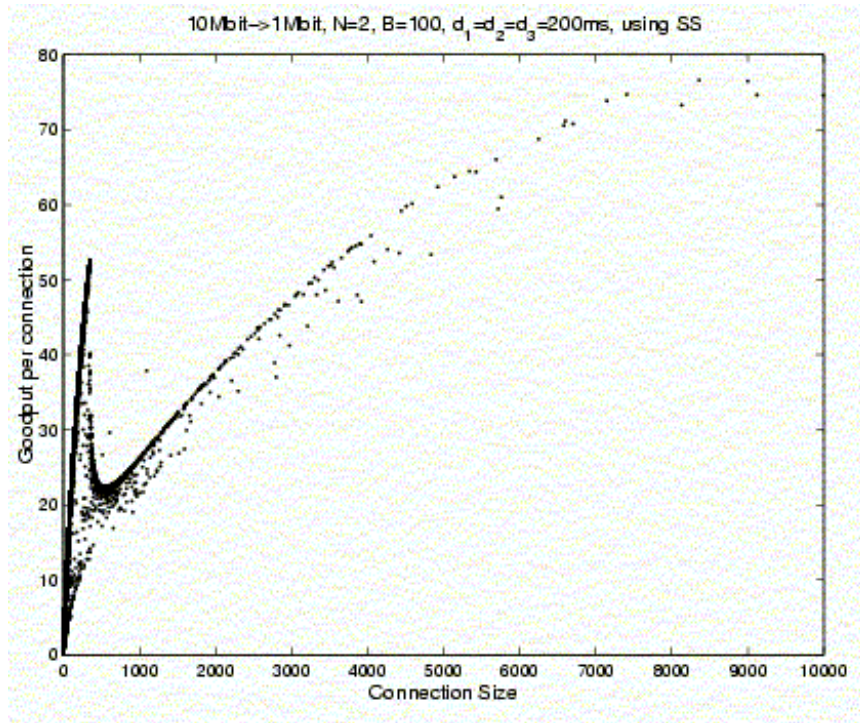


Simulation Model: Congestion Scenario with Multiplexed ON/OFF Traffic

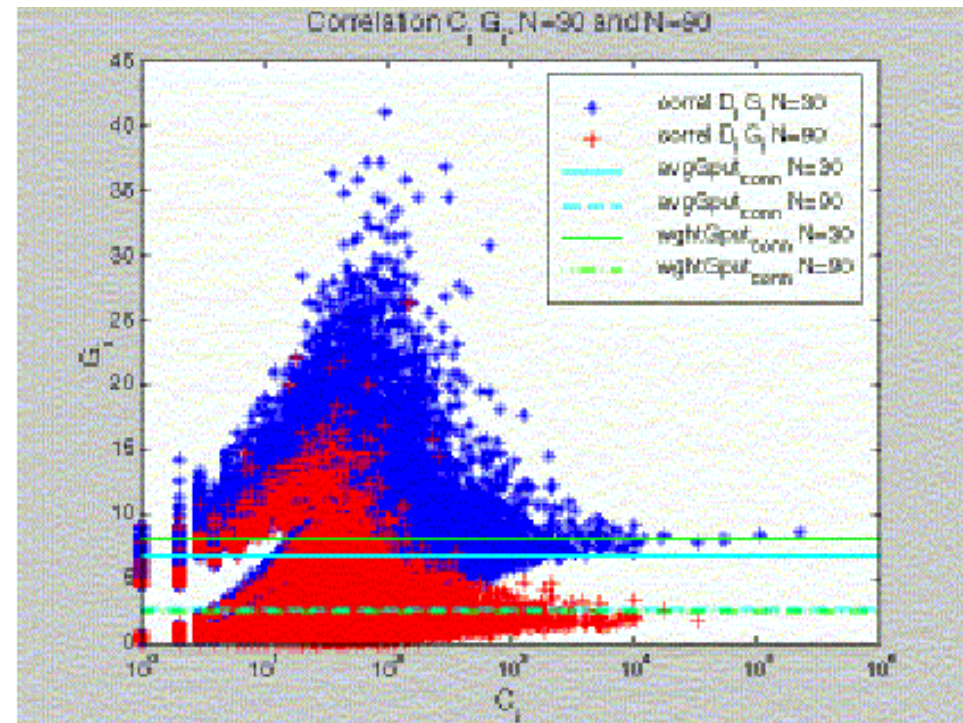


- Scenario: N ON/OFF sources, queueing/loss only at bottleneck router
- Here: Investigate average throughput per connection in TCP setting

Correlation: Connection Size $\leftrightarrow$ Throughput

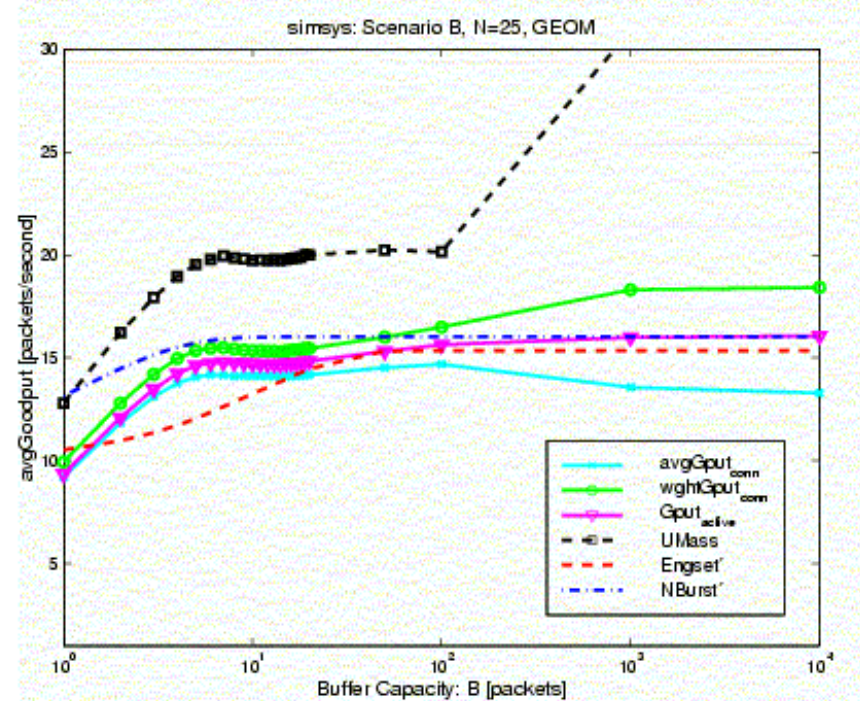
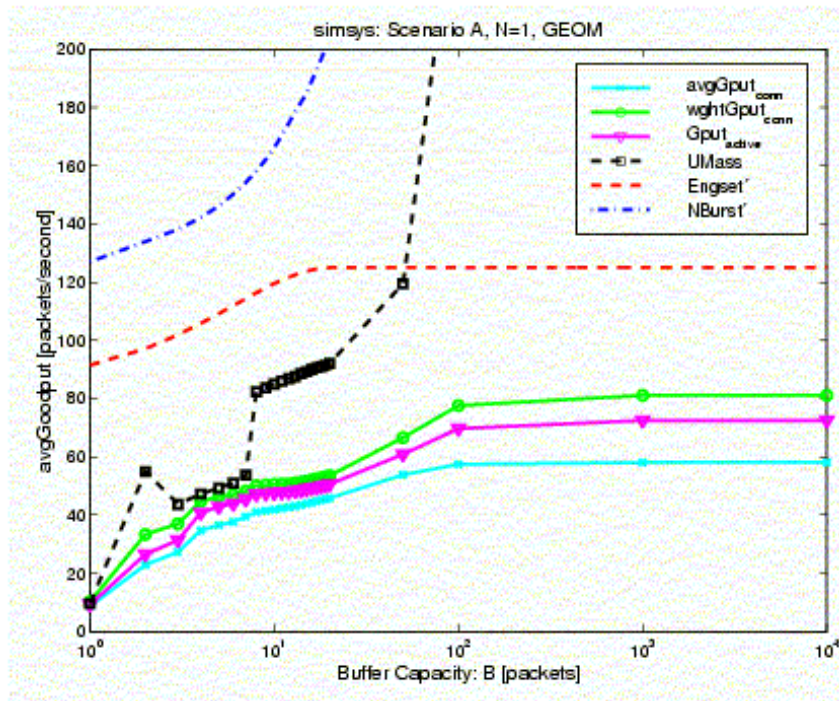


- $N=2$ bursty TCP sources



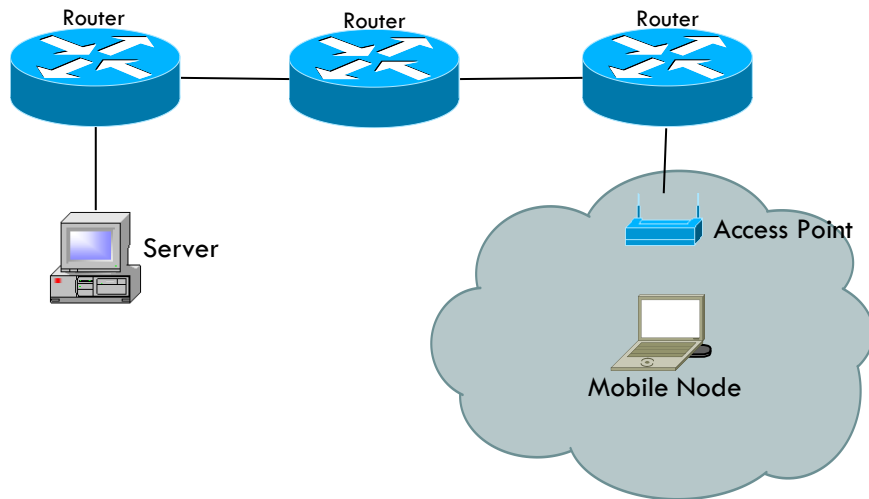
- $N=30$ (blue), and $N=90$ (red) TCP ON/OFF sources

Comparison: Models & simulation results



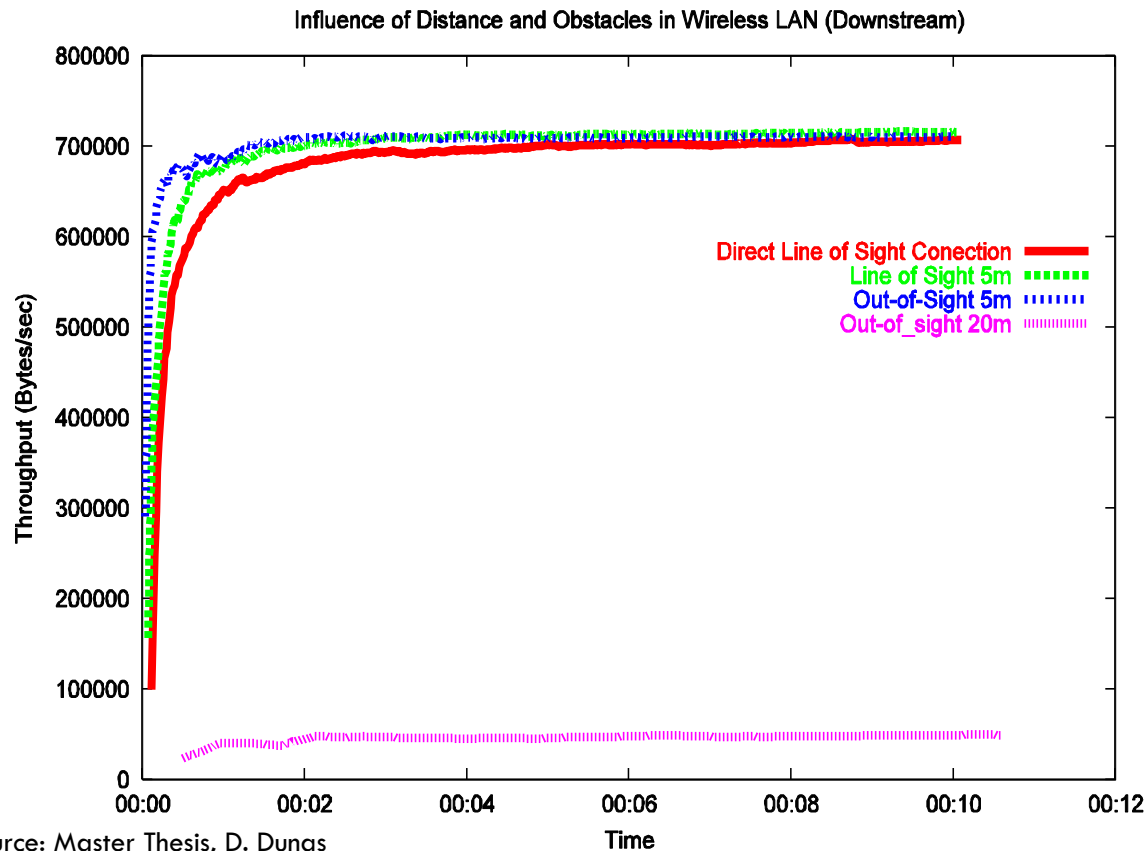
... see [ITCOM01] for details & more results

Measurements: Static Scenarios -- Network Topology



- 'persistent' TCP connection between mobile node and server
- Here: downstream data flow (server → mobile node)
- WLAN: 802.11b
 - ▣ Infrastructure mode
 - ▣ DCF access
- Bluetooth:
 - ▣ Access Point as Master
 - ▣ PAN profile
- Performance metric: (aggregated) throughput
 - ▣ $\Lambda(t) = V(t)/t$
 - ▣ $V(t)$ = received payload until time t

TCP over WLAN 802.11b

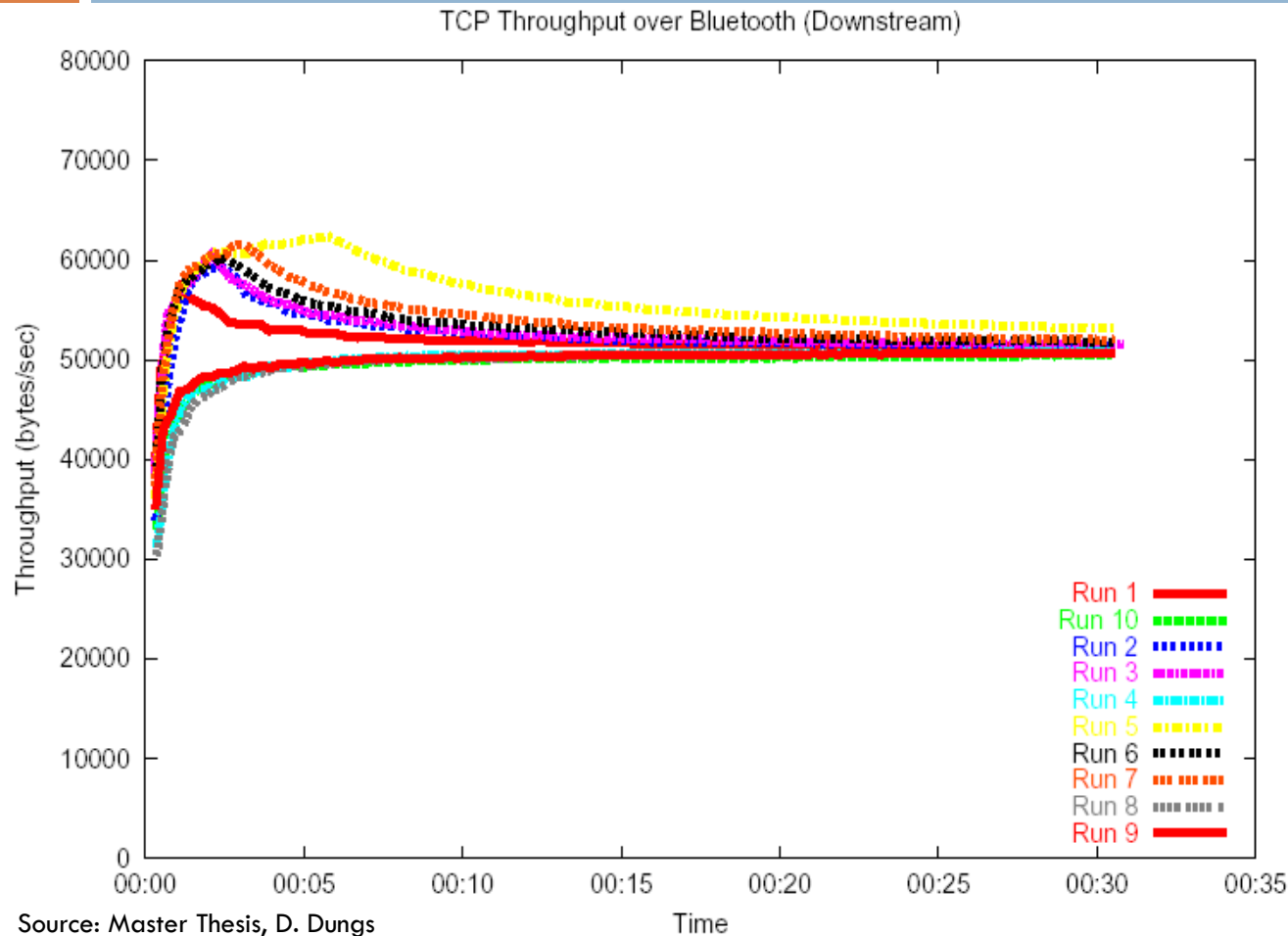


□ Strongly reduced throughput in scenario with poor channel conditions

□ No TCP retransmissions observed

→ WLAN link-layer retransmissions (together with rate adaptation) effective in these scenarios

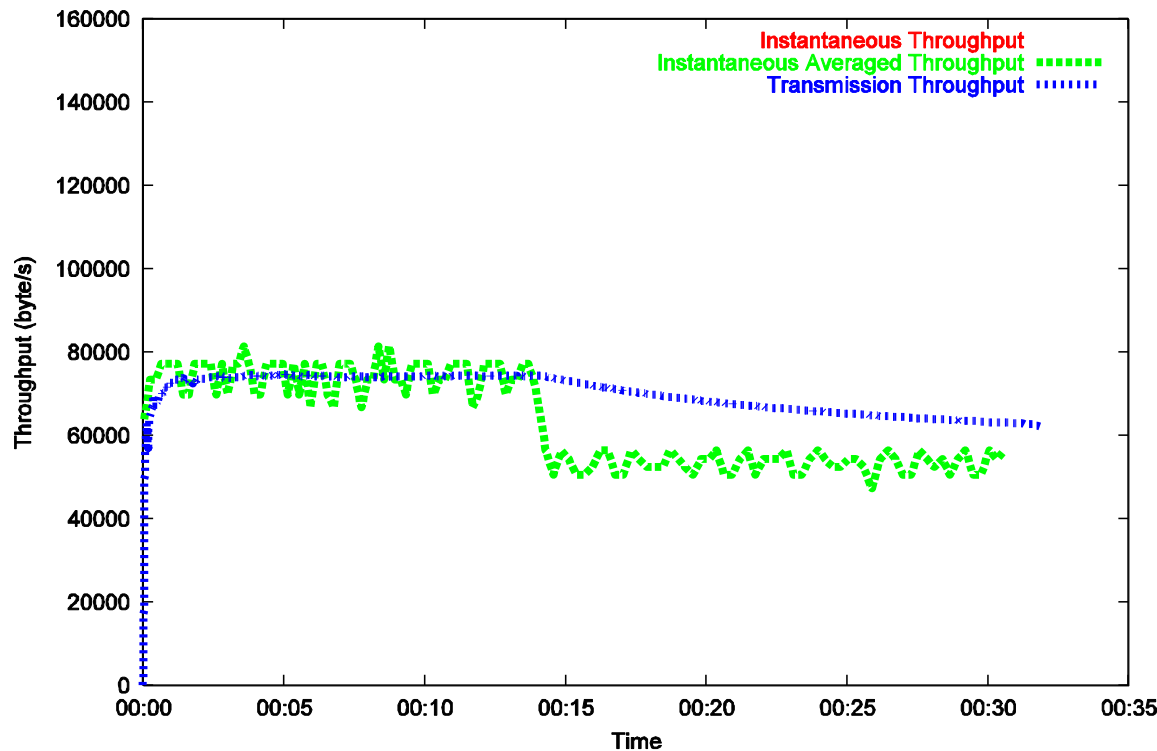
TCP over Bluetooth



- Long-term throughput smaller than achieved maximum in some measurements
- No TCP level retransmissions

TCP over Bluetooth: detailed analysis

UDP Throughput over time (Downstream)



- Additional performance metric: throughput average over 10 packets
- Throughput drop-down by approx. 33% after ca. 15 sec. in that experiment
- Possible explanation
 - BT Packet scheduler / link symmetry
 - [interference]

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21

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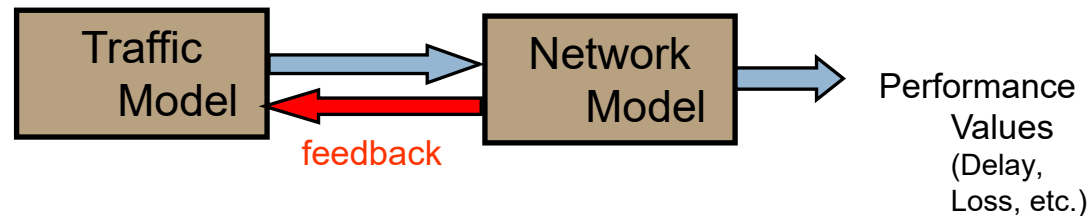
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Challenge: Traffic/Performance Models for TCP

End-to-End flow/congestion control

- Feedback: network behavior (congestion) $\leftrightarrow$ ingress traffic
- No separation traffic model/network model possible
- New traffic/performance models required



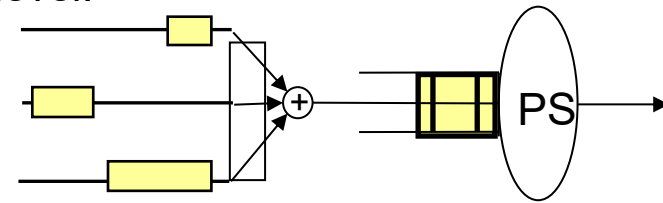
Approaches:

1. Connection level models (e.g. [Heyman et al., Sigmetrics 97])
2. Sender/receiver behavior & fixpoint iteration
3. Integrated models, e.g. multiplexed ON-OFF models

Connection Level model: e.g. [Heyman, Lakshman, Neidhardt]

Modified Processor Sharing model ($\sim M/M/1//N/PS$) on connection level:

- N TCP sources share bottleneck node (service rate ν)
- Maximum Transmission Rate λ_p within each connection
- For j active connections, each obtains service at rate
 - λ_p when $j\lambda_p \leq \nu$
 - $\delta\nu/j$ when $j\lambda_p > \nu$
- with attenuation factor $\delta \leq 1$ derived from TCP behavior

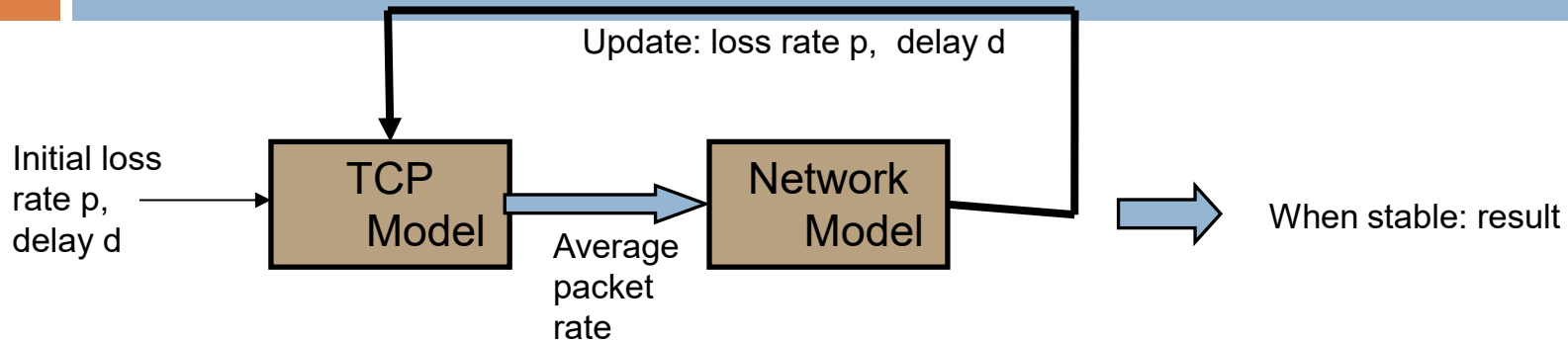


of customers in queues corresponds to number of active connections

$$\delta = 1 - \frac{[(1 - p_{low})^+]^2}{2(p_{high} - p_{low})} \quad p_{high} = \frac{B}{\nu R_0} + 1, \quad p_{low} = \frac{p_{high}}{2^{1.5}}$$

- R_0 = minimal Round-Trip Time
- B = Buffer Size at bottleneck

Fixpoint models



Network Model:

- E.g. M/M/1/K model or bulk-arrival queue

TCP model, e.g. [Padhye et al., Sigcomm 98]

$$T(p, \overline{RTT}) \approx \frac{1}{\overline{RTT}} \min \left(W_{max}, \frac{1}{\sqrt{\frac{2}{3}bp} + T_0/\overline{RTT} \min \left(1, 3\sqrt{\frac{3}{8}bp} \right) p(1 + 32p^2)} \right)$$

b: # packets acknowledged by ACK

RTT: Average Round-Trip-Time (including queuing delay)

W_{max} : Maximum size of congestion window

T_0 : Average Time-Out interval

p: Fraction of retransmitted packets

Model here is focused on RTT and losses: How to include throughput limitations from underlying Layer-2?

File Transfer delays with TCP: Models considering slow-start and link bandwidth limitations

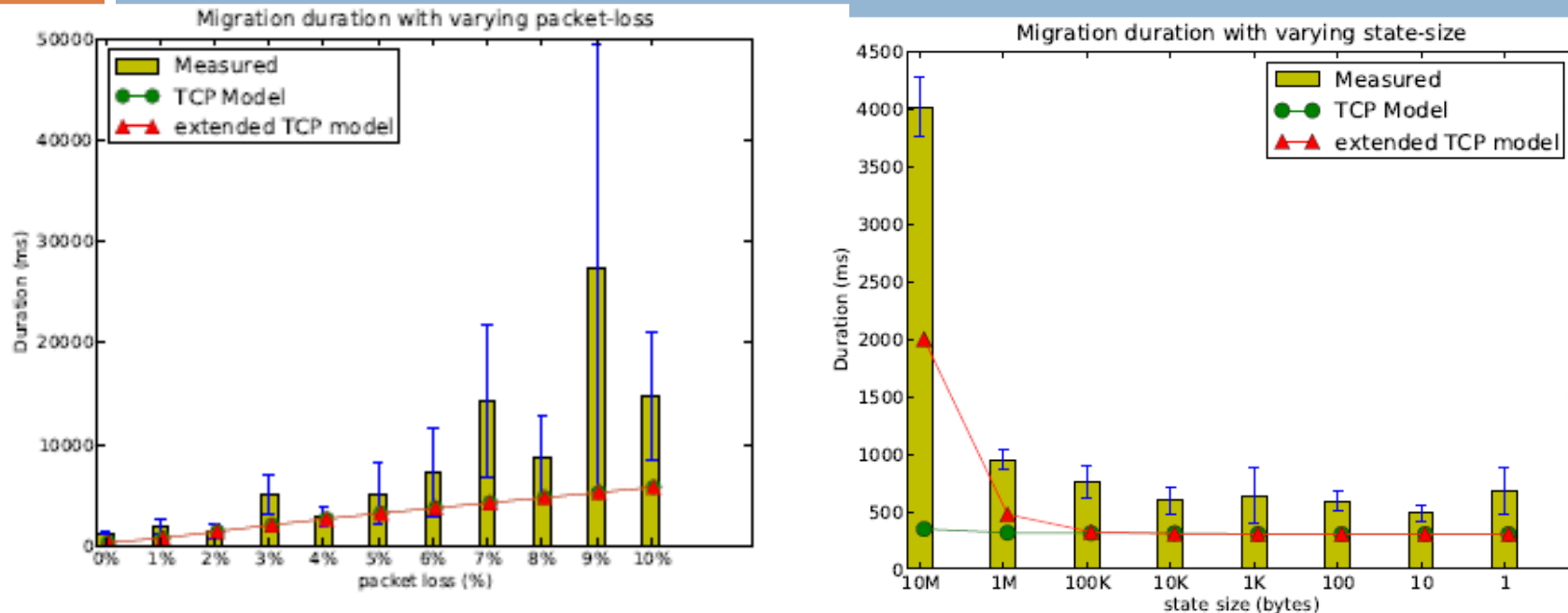


Fig. 9. Duration of a migration. Link packet loss is varied from 0 to 10%.

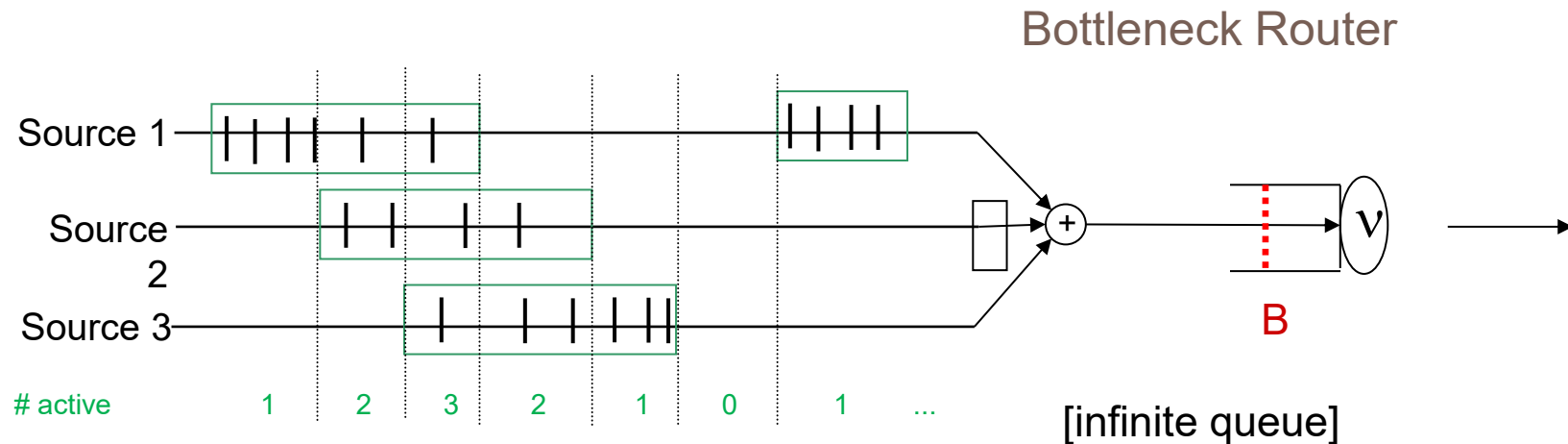
TCP Model from

[23] N. Cardwell, S. Savage, and T. Anderson, "Modeling tcp latency," vol. 3, mar. 2000, pp. 1742–1751 vol.3.

Extended model including bandwidth limitations,

see Højgaard-Hansen, Nguyen, Schwefel, Session mobility solution for client-based application migration scenarios', Proceedings of Wireless on Demand Networks 2011

Integrated Models: discouraged ON/OFF arrivals



Queueing Model with modified (queue-dependent) ON/OFF arrival process:

- Cmp. Processor sharing
- „Sharing“ of bandwidth: packet-rate λ_p at source reduced to $\delta v/j$ for j active sources when $j\lambda_p > v$
 - conn. duration extended $\rightarrow$ #packets in conn. unchanged
 - Throttling only during congestion state when queue-length $\geq B$
- $\rightarrow$ Markov Chain model structure?

Layer 4 Performance - Summary

27

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References

- TCP General - IETF RFC 9293 TCP - <https://www.ietf.org/rfc/rfc9293.html>
- Congestion Control : van Jacobsen, M. Karels – Congestion Avoidance and Control, Nov 1988, <https://ee.lbl.gov/papers/congavoid.pdf>, or JACOBSON , V. Congestion avoidance and control. In Proceedings of SIGCOMM '88 (Stanford, CA, Aug. 1988), ACM.
- Schwefel, Hans & Jobmann, Manfred & Hoellisch, Daniel & Heyman, Daniel. (2001). ITCOM 2001: International Symposium on the Convergence of IT and Communications. 91-102. 10.1117/12.434338.
- **Padhye, Firoiu, Towsley, Kurose: Modeling TCP throughput: a simple model and its empirical validation, ACM SIGCOMM Computer Communication Review Vol. 28, Iss 4, 1998.**
<https://dl.acm.org/doi/10.1145/285243.285291>
- Højgaard-Hansen, K, Schwefel, H-P & Nguyen, HC 2011, Session mobility solution for client-based application migration scenarios. in *Proceedings of Wireless On-Demand Network Systems and Services (WONS), 2011 Eighth International Conference on* . IEEE Press, pp. 76-83.
<https://doi.org/10.1109/WONS.2011.5720203>
- H. Schwefel, "Behavior of TCP-like elastic traffic at a buffered bottleneck router," *Proceedings IEEE INFOCOM 2001. Conference on Computer Communications. Twentieth Annual Joint Conference of the IEEE Computer and Communications Society (Cat. No.01CH37213)*, Anchorage, AK, USA, 2001, pp. 1698-1705 vol.3, doi: 10.1109/INFCOM.2001.916667.

Exercises – TCP Performance

29

- 1) File download via TCP: Select files of three different sizes from the Internet (say few kB, some 100 kB, few MB) that can be accessed via FTP or http (both use TCP as transport layer). Use any L2-connection type (e.g. WLAN, 2G, 4G, Ethernet) you prefer but keep it fixed during the exercise
 - a) Measure file download times – do 5 repetitions each and obtain a mean estimate and a confidence interval. Plot the behavior of the mean download time over size.
 - b) Measure the RTT to the IP address that hosts the FTP/HTTP server using PING. Obtain a mean estimate and confidence interval for that mean estimate.
 - c) Use the TCP throughput calculation of Slide 24 and compare it to your obtained results from Exercise 1a for different values of p . Make assumptions for the other parameters in that formula. Discuss the observed differences.
 - d) Use an $M/M/1/5$ model as network model. Choose some service rate that you find appropriate (and give an argument why). Make 3-4 fixpoint iterations of the model from Slide 24 (with your selected parameters) and see how the model behaves – does it converge? If so, to what value? If not, what would be the ‘best’ solution for the fixpoint problem?